

Keerthanaraj C H

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SUMMARY

Final-year Integrated M.Sc. Computer Science student specializing in Artificial Intelligence and Machine Learning. Skilled in Python, Deep Learning, Web Development, and Data Analysis. Experienced in developing and deploying ML-based applications using Streamlit and Django. Passionate about applying AI-driven solutions to real-world problems. Seeking an internship opportunity to contribute to innovative projects and gain hands-on industry experience.

EDUCATION

Nehru Arts and Science College, Kanhangad <i>Integrated M.Sc. Computer Science – AI & Machine Learning</i> CGPA (up to 9th Semester): 9.16	Kannur University 2021 – 2026
Government Higher Secondary School, Munderi <i>Higher Secondary – Board of Higher Secondary Examination, Kerala</i> Percentage: 98.5%	2021
Government Higher Secondary School, Munderi <i>Matriculation – Board of Public Examination, Kerala</i> Percentage: 98%	2019

ACHIEVEMENTS & EXAMINATIONS

UGC NET Qualified – Computer Science: Eligible for Assistant Professor, UGC (December 2025)
Ph.D. Admission Qualified through UGC NET (Computer Science and Applications) (July 2025)
Best Volunteer Award – National Service Scheme Unit, Nehru Arts and Science College

PROJECTS

- SignBridge: Bidirectional Explainable GNN for Indian Sign Language** | *Final Year Project* March 2026
- Built a real-time bidirectional ISL system with two modes: **Sign-to-Text-and-Speech** (webcam → GAT → voice) and **Text-to-Sign-Animation** (typed word → animated 3D hand skeleton).
 - Modelled hand joints as a graph (42 nodes, anatomical edges) using MediaPipe Hands; engineered 9-dimensional node features including wrist-normalised 3D coordinates and geometric distance features.
 - Trained and compared **9 models** (KNN, SVM, RF, XGBoost, ANN, DNN, CNN, ResNet, GAT); the proposed **Graph Attention Network achieved 79.76% accuracy**, outperforming all baselines.
 - Integrated **GNNExplainer** for post-hoc XAI, generating anatomically meaningful edge importance masks to highlight the joints most critical to each prediction.
 - Compiled a custom ISL dataset of **14 sign classes** × **~1,200 images** using Google Teachable Machine; deployed the full system as an **offline Streamlit application** with no internet dependency.
 - Technologies:** Python, PyTorch, PyTorch Geometric, MediaPipe, OpenCV, Scikit-learn, TensorFlow/Keras, XGBoost, Streamlit, Plotly, pytsx3, Matplotlib
- ISL Gesture Recognition & Speech Conversion** | *Deep Learning*
- Developed an earlier real-time system to recognize static ISL alphabets and digits and convert them into text and speech using CNN-based classification.
 - Integrated MediaPipe for hand-landmark extraction and deployed a webcam-based application with audio output via gTTS.
 - Technologies:** Python, TensorFlow/Keras, MediaPipe, OpenCV, NumPy, Scikit-learn, gTTS
- Food Nutrition Prediction & Meal Time Classification** | *Machine Learning – Case Study*
- Built ML models to predict kilocalories using Ridge Regression and classify meal times using Random Forest Classifier.
 - Preprocessed a curated Indian food dataset, applied feature selection, and created a Streamlit web app for real-time predictions.
 - Technologies:** Python, Streamlit, Scikit-learn, Pandas, NumPy, Matplotlib
- National Service Scheme Management System (NSSMS)** | *Web Application*
- Designed a digital platform to manage NSS activities across multiple user roles: Admin, Programme Officer, and Volunteer Secretary.
 - Enabled secure role-based access, automated event approvals, and volunteer tracking; reduced manual workload by **40%**.
 - Technologies:** Python, Django, MySQL, HTML, CSS, Bootstrap

TECHNICAL SKILLS

Programming Languages: Python, Java, C

Web Development: Django, Streamlit, HTML, CSS, Bootstrap

ML & Deep Learning: PyTorch, PyTorch Geometric, TensorFlow/Keras, Scikit-learn, XGBoost, GAT, GNNExplainer

Computer Vision: MediaPipe, OpenCV, Pandas, NumPy, Matplotlib, Plotly

Version Control & Tools: Git, GitHub, VS Code, Google Colab

Databases: MySQL

Core Skills: Problem Solving, Communication, Team Leadership, Critical Thinking

CERTIFICATIONS & WORKSHOPS

Programming for Everybody (Getting Started with Python) – Coursera, University of Michigan

Introduction to Artificial Intelligence – LinkedIn Learning

Generative AI – Microsoft & LinkedIn Learning

Ethics in the Age of Generative AI – LinkedIn Learning

Machine Learning Introduction (3-Day Workshop) – FOSS Club, Central University of Periya

Cloud Computing (One-Day Workshop) – IBM SkillsBuild & ASAP Kerala, Edunet Foundation

INTERESTS

Artificial Intelligence • Graph Neural Networks • Explainable AI • Data Science • Web Application Development • Generative AI